Building a Beam-Class LLM

The opportunity to develop a new open model that competes with Anthropic, OpenAI, Google, xAI and Meta

Executive summary · 19 October 2026 · Companion to the Research Baseline v2.0

The answer up front

30 vs 58

Beam vs. Claude Opus 5.5 on the independent Artificial Analysis Intelligence Index

$11–175M

Indicative 2-year cost to field a Beam-class model: adapt Beam vs. build from scratch

11%

Open-weight share of enterprise LLM API spend, down from 19% a year earlier

Verdict: a Beam-class model is buildable and affordable — but it will not beat Anthropic, OpenAI, Google or xAI on general capability. Win where closed models cannot follow: sovereignty, on-prem control, domain specialisation and cost. Start by adapting Beam's Apache 2.0 weights; gate any from-scratch build on traction.

Beam 1.0: what Reflection AI shipped

501B
total parameters

23.4B
active per token

256 / 8
experts / active

1M
token context

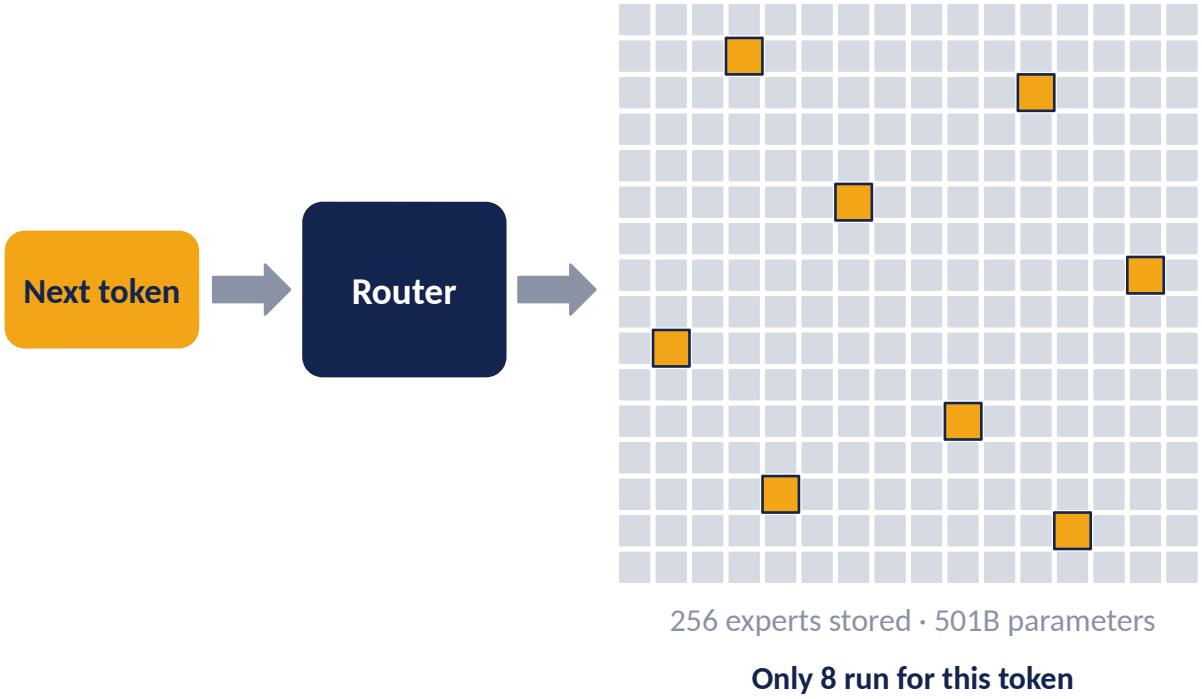
23.8T
pre-training tokens

Apache 2.0
open weights, 16 Oct

Timeline

- 2024 · founded by ex-DeepMind team (Laskin, Antonoglou)
- Oct 2025 · $2B raise led by NVIDIA
- 5 Oct 2026 · Beam announced
- 16 Oct 2026 · weights, Mirror CLI harness and DFlash2 drafter released; 40+ partners
- Independent score: 30 (19th of 120)

Architecture built for cheap tokens



1 A router reads the token
and scores every expert

2 256 specialist experts
hold the model's knowledge

3 Only the top 8 wake up
23.4B of 501B parameters (4.7%)

Compute per token

Dense 501B model

Beam (4.7%)

≈ 21× less compute

Why each token costs less

1 · Sparse experts

Only 8 of 256 experts run per token, so a 501B model computes like a ~23B one.

21×

less compute

2 · Short memory in most layers

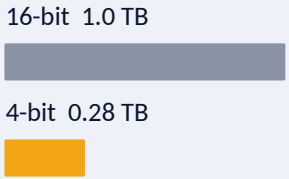
43 of 52 layers only look at the last 512 tokens; just 9 read the whole context. Memory ≈ 17% of full attention.



512-token window slides along

3 · 4-bit weights (NVFP4)

Weights shrink to ~0.28 TB and fit one 8-GPU server, losing ≤ 1.7 points on agentic benchmarks.



4 · Fewer, faster tokens

A 1.9B “drafter” guesses several tokens that Beam checks in one pass, and Beam uses ~40% fewer output tokens per task.

−40%

output tokens

The trade-off: frontier-scale knowledge at small-model running cost, but capability still trails the frontier (independent index 30 vs 58).

Training: post-training is the big bill

1

Curate

~95% of web text discarded

2

Pre-train

23.8T tokens · 6,144 GB300 · <4 wks

3

Mid-train

Context to 1M tokens

4

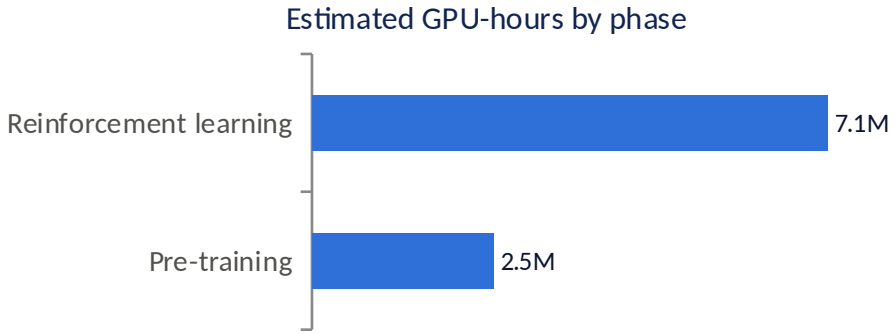
SFT

Tool use & agent traces

5

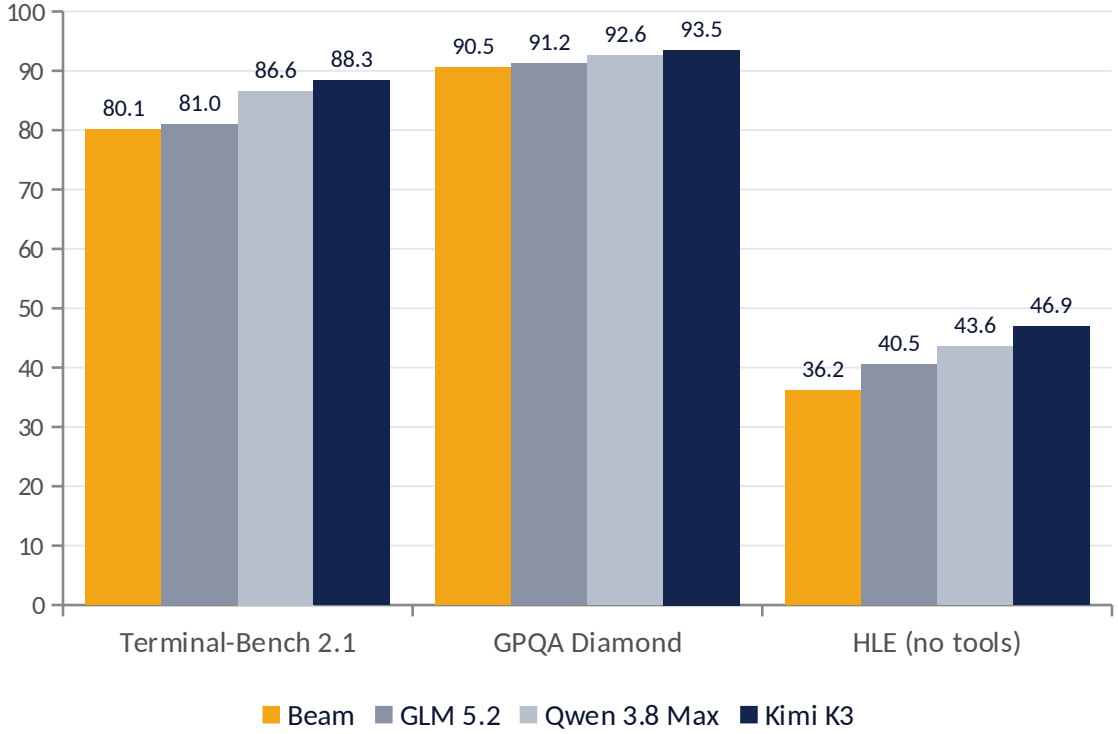
RL

10,500 GB300 · 4 wks · 100M+ rollouts



RL used ~2–3× the GPU-hours of pre-training. Budget for an environment fleet (~1M tasks, ~1.3B sandboxes), not just a training cluster.

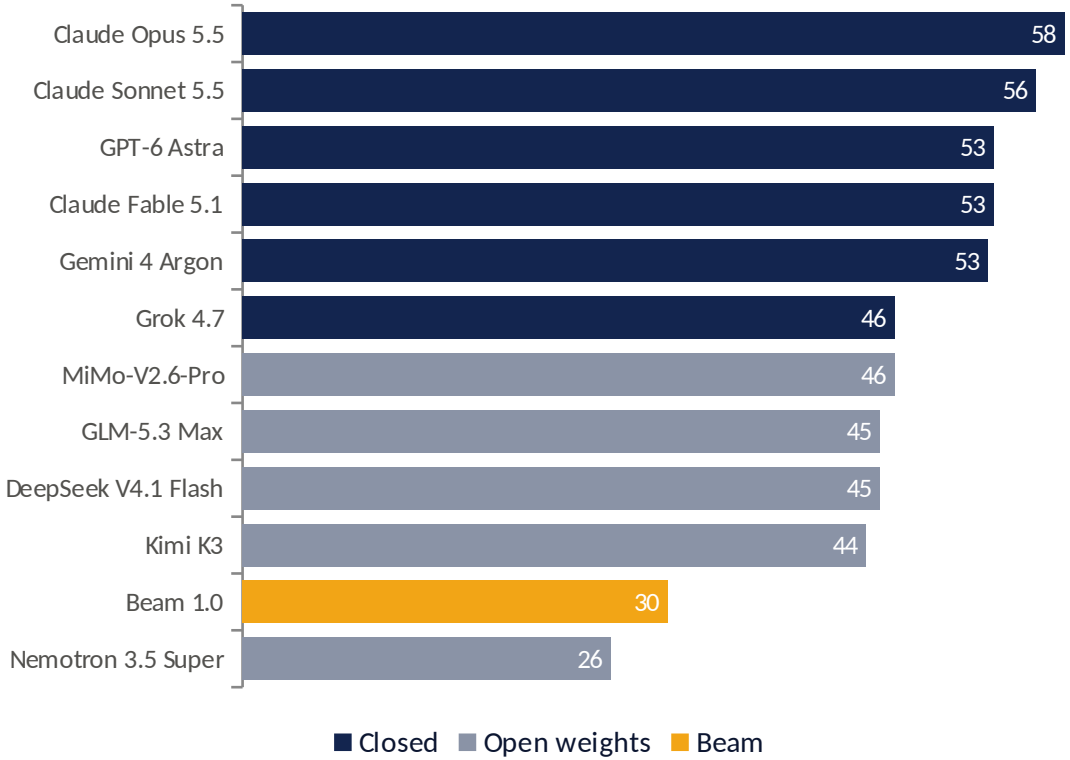
On its own numbers, Beam trails open peers



Reflection's launch table (vendor-reported)
Beam is roughly level with GLM-5.2 on agentic coding and science QA, but behind Kimi K3 and Qwen 3.8 Max on every shared benchmark.

Its pitch is efficiency: similar quality with 23B active parameters vs. 40–95B for peers.

Independent tests: a 28-point gap



Artificial Analysis Intelligence Index, Oct 2026

One scale, one harness, 120 models.

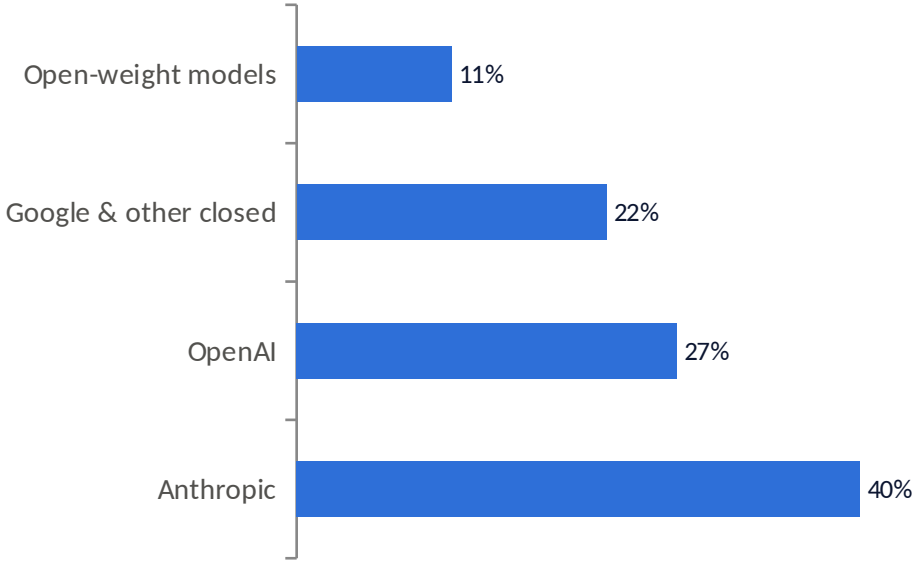
- Closed frontier: 53–58
- Grok and top Chinese open models: 44–46
- Beam: 30, but ~40% fewer output tokens than the median

The vendor landscape

Vendor	Flagship	Weights	AA Index	Output $/1M	Edge
Anthropic	Claude Opus 5.5	Closed	58	$20	Top capability; 40% enterprise share
OpenAI	GPT-6 Astra	Closed	53	$50	Reach; reasoning
Google	Gemini 4 Argon	Closed	52.6	—	Multimodal; cloud; cheap Flash tier
xAI	Grok 4.7	Closed	46	$6	Price and speed
Meta	Muse Spark / Llama 4	Closed / open	—	—	Distribution; Llama brand
China open labs	MiMo · GLM · Kimi	Open	44–46	$0.28+	Best open scores; lowest prices
Reflection	**Beam 1.0**	Open	**30**	Partner	US provenance; efficiency

Weights "Open" for Reflection = Apache 2.0. Output prices per 1M tokens (aggregators, Oct 2026). Groq (inference hardware) hosts open models and is a deployment partner, not a model competitor.

Enterprise spend favours closed models



$37B

enterprise gen-AI spend in 2025 (≈3× y/y)

19% → 11%

open-weight share in one year

Source: Menlo Ventures, late 2025. "Google & other closed" is the remainder.

The market is growing fast, but buyers pay for the most capable closed models. An open model has to offer something closed APIs cannot.

Where a new open model can win

Sovereign & government

Allies that cannot procure Chinese weights and need on-prem control

Regulated industries

Finance, health, defence, legal: residency and audit beat a few index points

Vertical specialists

Post-train on proprietary environments to beat general models on narrow tasks

High-volume agents

Coding and automation agents where cost per token dominates

Under-served languages

Hebrew, Arabic, regional European languages via targeted data

Customisation

Full fine-tuning, RL on customer tasks, predictable pricing

Three levels of ambition

Level 1

Beam parity

AA ≈ 30

~500B total / 23B active; Beam KPIs

2-yr cost: $11M–$175M

Level 2

Open-frontier parity

AA ≈ 45

1–2T total / 40–60B active; 30T+ tokens

2-yr cost: $300–500M

Level 3

Frontier challenger

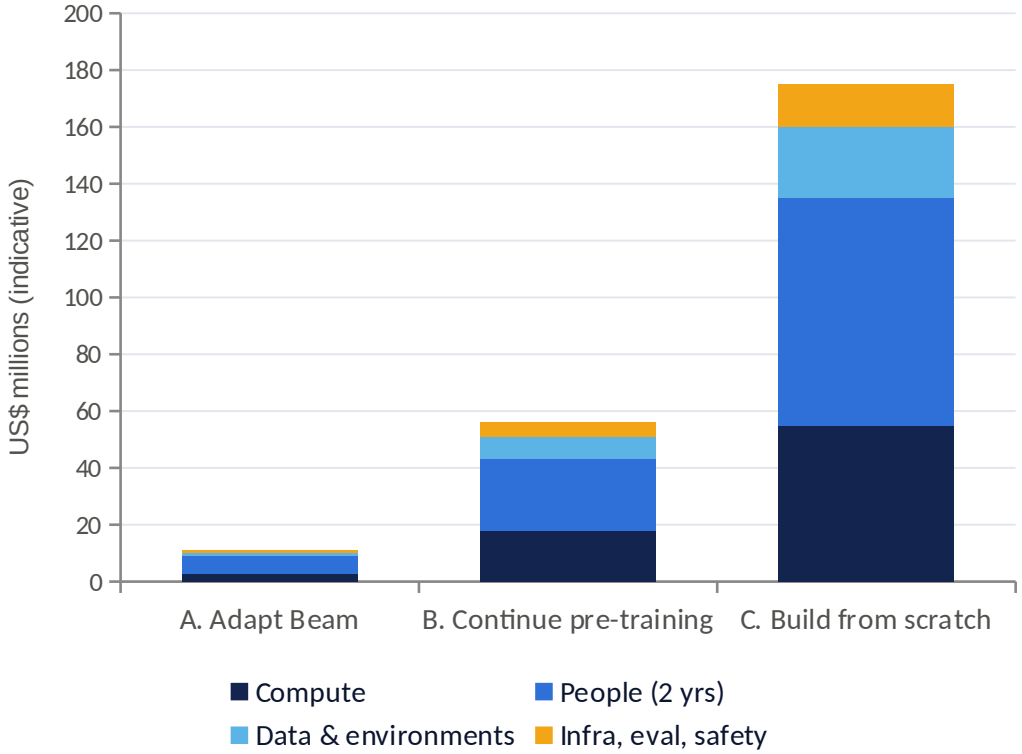
AA ≥ 53

Multi-trillion params; 10^25–10^26 FLOPs

2-yr cost: $1B+ per generation

Recommendation: commit to Level 1 now; treat Level 2 as a funded option after proof of demand. Level 3 is a hyperscaler-scale bet.

Three routes to a Beam-class model



A · ≈$11M · 3–6 months

SFT + RL on Beam weights for a niche; 15–25 people

B · ≈$56M · 9–12 months

Add 3–6T domain tokens + full RL; 40–60 people

C · ≈$175M · 15–18 months

Own model, ~11–14M GPU-hours; 80–120 people

Staged roadmap with decision gates



Reproduce Beam & peers on our own harness	Adapted Beam +5 pts on domain evals; first paying pilot	Pilot model hits scaling targets; 20T tokens; ≥$150M funding	Stable full training run under way	Level 1 KPIs met independently; safety report; partners

G2 is the key investment decision: only proceed to a from-scratch model if Stage 1 shows demand and funding is secured.

Key risks and how we manage them

Frontier keeps moving

Don’t race the leaderboard; own a niche

Chinese open models at near-zero price

Sell provenance, sovereignty, support

Claims that don’t replicate

Independent harness from day one

GPU cost and availability

Multi-provider reservations; NVFP4

RL infrastructure underestimated

Build environments from month 1

Talent and legal exposure

Equity, partnerships; provenance tracking

Recommendation and next steps

1

Choose the wedge

Pick one or two segments: sovereign, regulated, vertical or language

2

Adapt Beam (Stage 1)

≈$11M, 3–6 months: SFT + RL on Beam 1.0 for that wedge

3

Build the hard parts early

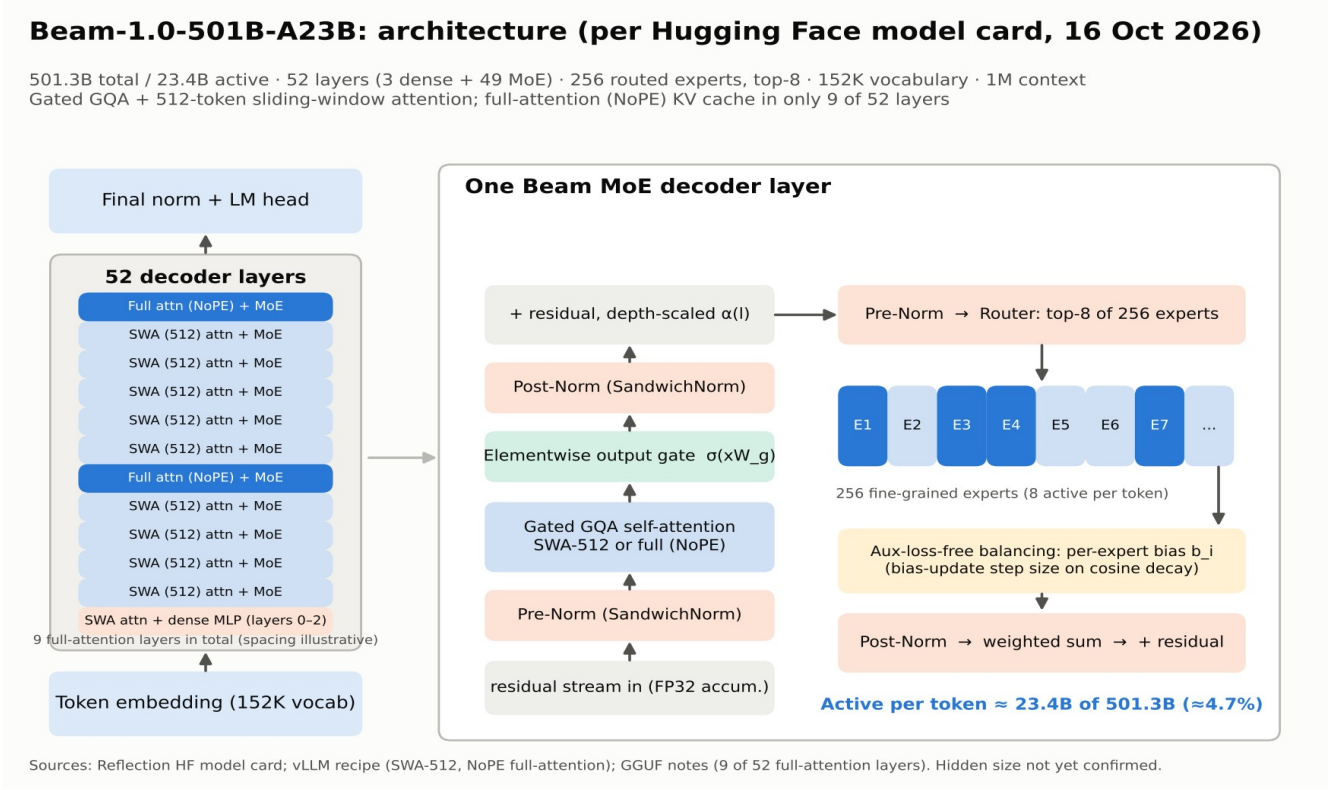
Data engine, RL environment fleet, independent eval harness

4

Decide at Gate 2

Fund continued pre-training or a from-scratch model only on proven demand

Appendix: Beam architecture detail



Sources and notes

- Reflection AI: Introducing Beam (5 Oct 2026); Beam is now available (16 Oct 2026)
- Hugging Face model cards: reflection/Beam-1.0-501B-A23B (BF16, FP8, NVFP4); vLLM and NVIDIA serving recipes
- Artificial Analysis Intelligence Index v4.3.x and Beam article (Oct 2026)
- Vendor launch materials and system cards: Anthropic (Claude Opus 5.5, Sonnet 5.5, Fable 5.1), OpenAI (GPT-6), Google (Gemini), xAI (Grok 4.7), Meta (Muse, Llama 4)
- Menlo Ventures enterprise LLM market data; pricing aggregators (BenchLM, Oct 2026)
- Costs, GPU-hours and team sizes are author estimates; see Sections 12 and Appendix A of the research baseline

Caveat: vendors report different benchmark versions and harnesses. Only the Artificial Analysis index puts all models on one scale; treat cross-vendor gaps under ~2 points as noise. Full source list (42 references) in the research baseline document.